Informatics Practices Practice Worksheet

Section B – 2 Mark Questions

1. SQL String & Date Functions

Practice 1: Write SQL queries to perform the following tasks:

(i) Extract the leftmost 4 characters after removing the leading and trailing spaces from " DATA SCIENCE ".

(ii) Display the day name of the week when India celebrated its Republic Day first, i.e., '26-01-1950'.

Practice 2: Write SQL queries to perform the following tasks:

(i) Extract the rightmost 3 characters after removing the leading and trailing spaces from " CYBER SECURITY ".

(ii) Display the month name of the date when World Environment Day is celebrated, i.e., '05-06-2024'.

2. Networking Concepts

Practice 1:
(a) What is a Web Server?

OR

(b) Write any one advantage and one disadvantage of Star topology.

Practice 2:
(a) What is a Web Browser?

OR

(b) Write any one advantage and one disadvantage of Mesh topology.

3. SQL Aggregate Functions

Practice 1: What are aggregate functions in SQL? Name any two aggregate functions.

Practice 2: Differentiate between single-row functions and multiple-row (aggregate) functions in SQL. Give one example of each.

4. NumPy to Pandas Series

Practice 1: Write a Python program to create a NumPy array containing the names of five vegetables. Convert it into a Pandas Series with custom indices ['V1', 'V2', 'V3', 'V4', 'V5'].

Practice 2: Write a Python program to create a NumPy array containing the names of five colors. Convert it into a Pandas Series with custom indices ['C1', 'C2', 'C3', 'C4', 'C5'].

5. Pandas Series Operations (Addition/Subtraction)

Practice 1: Write the output of the following code:

```
import pandas as pd
s1 = pd.Series([10, 20, 30], index=['x', 'y', 'z'])
s2 = pd.Series([5, 15, 25], index=['y', 'z', 'w'])
print(s1 + s2)
```

Practice 2: Write the output of the following code:

```
import pandas as pd
s1 = pd.Series([100, 200, 300], index=['A', 'B', 'C'])
s2 = pd.Series([50, 150, 250], index=['B', 'C', 'D'])
print(s1 - s2)
```

6. SQL String Functions

Practice 1: Consider the string: "Information Technology". Write suitable SQL queries for the following:

I. To extract and display "format" from the string.

II. Display the position of the first occurrence of "Tech" in the given string.

Practice 2: Consider the string: "Computer Science". Write suitable SQL queries for the following:

I. To extract and display "puter" from the string.

II. Display the position of the first occurrence of "Sci" in the given string.

7. Pandas Code Completion

Practice 1: Rohan wants to create a Pandas Series using a NumPy array with three elements 'x', 'y', and 'z', and assign custom indexes 'one', 'two' and 'three'. Help him complete the code below.

```
import pandas as pd
import _________ as np
arr = np._________(['x', 'y', 'z'])
s = pd._________(arr, _________ = ['one', 'two', 'three'])
print(s)
```

Practice 2: Priya wants to create a Pandas Series using a NumPy array with three elements 10, 20, and 30, and assign custom indexes 'r1', 'r2' and 'r3'. Help her complete the code below.

```
import pandas as pd
import _________ as np
arr = np._________([10, 20, 30])
s = pd._________(arr, _________ = ['r1', 'r2', 'r3'])
print(s)
```

Section C – 3 Mark Questions

8. Pandas Series from Dictionary

Practice 1: Write a Python program to create a Pandas Series as shown below using a dictionary. Note that the left column indicates the indices and the right column displays the data.

Index	Data
Japan	Tokyo
France	Paris
Italy	Rome

Practice 2: Write a Python program to create a Pandas Series as shown below using a dictionary. Note that the left column indicates the indices and the right column displays the data.

Index	Data
Australia	Canberra
Canada	Ottawa
Germany	Berlin

9. SQL Table Creation and Insertion

Practice 1: (i) Write an SQL statement to create a table named EMPLOYEES, with the following specifications:

Column Name	Data Type	Key
EmpID	Numeric	Primary Key
FirstName	Varchar(30)	
LastName	Varchar(20)	
HireDate	Date	
Salary	Float(10,2)	

(ii) Write an SQL query to insert the following data in the EMPLOYEES table:
1, Rahul, Sharma, 2015-09-12, 55000.00

Practice 2: (i) Write an SQL statement to create a table named PRODUCTS, with the following specifications:

Column Name	Data Type	Key
ProductID	Numeric	Primary Key
ProductName	Varchar(40)	
Category	Varchar(20)	
MfgDate	Date	
Price	Float(8,2)	

(ii) Write an SQL query to insert the following data in the PRODUCTS table:
101, Laptop, Electronics, 2023-01-15, 75000.50

10. Pandas Series from Scalars

Practice 1: Write a Python program to create and display a Pandas Series as shown below:

Month	Days
Jan	31
Feb	28
Mar	31
Apr	30

The index labels are the names of the months - Jan, Feb, Mar, Apr. The corresponding scalar values are the number of days in these respective months - 31, 28, 31, 30.

Practice 2: Write a Python program to create and display a Pandas Series as shown below:

Quarter	Months
Q1	3
Q2	3
Q3	3
Q4	3

The index labels are the quarter names - Q1, Q2, Q3, Q4. The corresponding scalar values are the number of months in these respective quarters - 3, 3, 3, 3.

11. Pandas Series from a Single Scalar

Practice 1: Write a Python program to create a Pandas Series from a scalar value. The Series should have indexes as 'i1', 'i2', 'i3', 'i4' with each value in the Series set to 500.

Practice 2: Write a Python program to create a Pandas Series from a scalar value. The Series should have indexes as 'p', 'q', 'r', 's', 't' with each value in the Series set to 2025.

Section D – 4 Mark Questions

12. SQL Query Output (Table Based)

Practice 1: Raghav, who works as a database designer, has created a table Employee as shown below:

EmpID	Name	City	Salary	Joining_Date
201	Anil Kumar	Chennai	45000	2020-01-15
202	Meena Joshi	Pune	55000	2019-07-20
203	Ravi Teja	Hyderabad	60000	2021-03-10
204	Sneha Rao	Chennai	48000	2022-05-05
205	Karan Singh	Delhi	40000	2018-11-01

Write the output of the following SQL queries:

(i) SELECT UPPER(Name) FROM Employee WHERE City = 'Chennai';

(ii) SELECT Name FROM Employee WHERE MONTH(Joining_Date) = 3;

(iii) SELECT Name FROM Employee WHERE Salary > 50000;

(iv) SELECT COUNT(*) FROM Employee WHERE City = 'Pune';

Practice 2: Rohit, who manages a library, has created a table BOOKS as shown below:

BookID	Title	Author	Price	Pub_Date
B001	Python Basics	Gupta	450	2021-04-12
B002	Data Science	Sharma	800	2020-08-25
B003	SQL Guide	Verma	300	2022-01-10
B004	AI Fundamentals	Gupta	950	2023-06-15
B005	Cloud Computing	Nair	600	2021-12-05

Write the output of the following SQL queries:

(i) SELECT LOWER(Author) FROM BOOKS WHERE Price > 600;

(ii) SELECT Title FROM BOOKS WHERE YEAR(Pub_Date) = 2021;

(iii) SELECT Title FROM BOOKS WHERE Price < 500;

(iv) SELECT COUNT(*) FROM BOOKS WHERE Author = 'Gupta';

13. SQL Query Output (Inventory Table)

Practice 1: A database table STATIONERY is given below:

ITEM_ID	ITEM_NAME	CATEGORY	QTY	PUR_DATE
S001	BALL PEN	WRITING	150	2024-01-15
S002	NOTEBOOK	OFFICE	75	2024-03-10
S003	MARKER	WRITING	40	2024-02-22
S004	FOLDER	OFFICE	200	2024-04-05
S005	GLUE	STATIONERY	30	2024-05-20

Write the output of the following SQL queries:

(i) Select UPPER(ITEM_NAME) from STATIONERY where CATEGORY = 'WRITING';

(ii) Select ITEM_NAME from STATIONERY where MONTH(PUR_DATE) = 4;

(iii) Select ITEM_NAME from STATIONERY where QTY < 50;

(iv) Select COUNT(*) from STATIONERY where CATEGORY = 'OFFICE';

Practice 2: A database table INVENTORY is given below:

P_ID	P_NAME	TYPE	STOCK	EXP_DATE
P101	MOUSE	ELECTRONICS	60	2025-06-10
P102	KEYBOARD	ELECTRONICS	25	2025-08-15
P103	PEN DRIVE	STORAGE	100	2026-01-20
P104	MONITOR	ELECTRONICS	15	2024-12-05
P105	HDD	STORAGE	45	2025-09-30

Write the output of the following SQL queries:

(i) Select LOWER(P_NAME) from INVENTORY where TYPE = 'ELECTRONICS';

(ii) Select P_NAME from INVENTORY where MONTH(EXP_DATE) = 1;

(iii) Select P_NAME from INVENTORY where STOCK < 50;

(iv) Select COUNT(*) from INVENTORY where TYPE = 'STORAGE';

Section E – 5 Mark Questions

14. Networking Case Study: "ABC Innovators Pvt. Ltd."

ABC Innovators Pvt. Ltd., Bengaluru is a company specializing in IoT solutions. They have different departments: **Research, Development, Marketing and IT Support**. The layout of the Bengaluru branch has the following distances between departments:

From	To	Distance
Research	Development	60 m
Research	Marketing	80 m
Research	IT Support	70 m
Development	Marketing	100 m
Development	IT Support	90 m
Marketing	IT Support	110 m

Department	No. of Computers
Research	40
Development	120
Marketing	100
IT Support	20

Questions:

(i) Suggest the topology and draw the most suitable cable layout for connecting all the departments in the Bengaluru office.

(ii) Suggest the type of network (LAN/MAN/WAN) that will be created when the Research department is connected to the Development department. Justify your answer.

(iii) Suggest the department for the placement of the server. Explain the reason for your selection.

(iv) Suggest the placement of switch/hub with justification.

(v) Employees of Bengaluru campus frequently use VoIP technology to communicate with their foreign clients. What is the full form of VoIP?

> **Help Notes for Practice:**
>
> - **(i) Topology & Layout:** Use **Star topology** for efficient management. Draw a minimum spanning tree starting from the central block (Development, as it has the most computers) and connect it to the nearest blocks.
> - **(ii) Type of Network:** Connecting Research to Development within the same campus creates a **LAN (Local Area Network)**, as it is confined to a limited geographical area (the office building).
> - **(iii) Server Placement:** Place the server in the **Development department** because it has the maximum number of computers (120), which centralizes data access and minimizes cabling to the majority of users.
> - **(iv) Switch/Hub Placement:** Place a switch/hub in **each department** to connect the computers locally, and connect these switches to a central main switch/router.
> - **(v) VoIP:** Full form is **Voice over Internet Protocol**.

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15. Networking Case Study: "ABC Pvt. Ltd."

ABC Pvt. Ltd., a multinational technology company, is looking to establish its Indian Head Office in Bengaluru, and a regional office branch in Lucknow. The Bengaluru head office will be organized into four departments: **HR, FINANCE, TECHNICAL, and SUPPORT**.

From	To	Distance
HR	Finance	65 m
HR	Technical	80 m
HR	Support	70 m
Finance	Technical	60 m
Finance	Support	75 m
Technical	Support	50 m

Bengaluru Office to Lucknow: 1900 km

Department/Office	No. of Computers
HR	175
Finance	35
Technical	50
Support	15
Lucknow Office	40

Questions:

(I) Suggest the most suitable department in the Bengaluru Office Setup, to install the server. Also, give a reason to justify your suggested location.

(II) Draw a suitable cable layout of wired network connectivity between the departments in the Bengaluru Office.

(III) Which networking device would you suggest the company to purchase to interconnect all the computers within a department in Bengaluru Office?

(IV) The company is considering establishing a network connection between its Bengaluru Head Office and Lucknow regional office. Which type of network - LAN, MAN, or WAN - will be created? Justify your answer.

(V) The company plans to develop an interactive website that will enable its employees to monitor their performance after login. Would you recommend a static or dynamic website, and why?

Help Notes for Practice:

- **(I) Server Placement:** Place the server in the **HR department**, as it has the maximum number of computers (175), minimizing overall cable length and providing efficient centralized management.
- **(II) Cable Layout:** Draw a minimum spanning tree. Connect HR to Finance (65m) and HR to Support (70m), then connect Technical to Finance (60m) or Support (50m) depending on the shortest path to cover all.
- **(III) Networking Device:** Use a **Switch** to interconnect all computers within a single department for better performance and collision avoidance.
- **(IV) Type of Network:** Connecting Bengaluru to Lucknow (1900 km) creates a **WAN (Wide Area Network)**, as it spans across large geographical distances (different cities).
- **(V) Website Type:** Recommend a **Dynamic website** because it needs to handle user logins and display real-time performance data fetched from a database, which static sites cannot do.

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16. SQL Queries (Functions & Conditions)

Practice 1:

(a) Write suitable SQL queries for the following:

(i) Display the maximum value in the Salary column from the EMPLOYEES table.

(ii) Display the FirstName column values in uppercase from the EMPLOYEES table.

(iii) Retrieve the DepartmentName column values in lowercase from the DEPARTMENTS table, where the DepartmentName has a length of 8 characters.

(iv) Show the remainder when dividing each value in the Salary column by 500 in the EMPLOYEES table.

(v) Count the rows in the EMPLOYEES table where Salary is less than 30000.

OR

(b) Write suitable SQL queries for the following:

(i) Find the power of 3 raised to 4.

(ii) Convert the string 'PYTHON PROGRAMMING' to lowercase.

(iii) Display the first 5 characters of the string 'Database'.

(iv) Extract the month from the date '2024-08-15'.

(v) Display the number of characters in the string 'Artificial Intelligence'.

Practice 2:

(a) Write suitable SQL queries for the following:

(i) Display the average value in the Marks column from the STUDENTS table.

(ii) Display the LastName column values in lowercase from the EMPLOYEES table.

(iii) Retrieve the ProductName column values in uppercase from the PRODUCTS table, where the ProductName has a length of 5 characters.

(iv) Show the remainder when dividing each value in the Price column by 100 in the PRODUCTS table.

(v) Count the rows in the STUDENTS table where Marks is greater than or equal to 90.

OR

(b) Write suitable SQL queries for the following:

(i) Find the power of 2 raised to 10.

(ii) Convert the string 'DATA SCIENCE' to uppercase.

(iii) Display the last 3 characters of the string 'Computer'.

(iv) Extract the year from the date '2025-12-31'.

(v) Display the number of characters in the string 'Informatics Practices'.

> ***General Tip for Cable Layout (Part ii of Q14/Q15):***
> *Draw a graph connecting all the blocks using the minimum total cable length. Find the block with the most connections (central block) and connect it to the nearest blocks, then connect the remaining blocks to the nearest already-connected block (Prim's/Kruskal's algorithm concept).*